



11567 - A High-Definition View of the Baryon Cycle in the First Two Billion Years

Cycle: 5, Proposal Category: GO

INVESTIGATORS

<i>Name</i>	<i>Institution</i>
Prof. Hsiao-Wen Chen (PI)	University of Chicago
Dr. Michael Rauch (CoI) (CoPI)	Carnegie Institution of Washington
Dr. Gwen C. Rudie (CoI) (CoPI)	Carnegie Institution of Washington
Dr. Gregory Walth (CoI)	California Institute of Technology
Rieko Momose (CoI)	Carnegie Institution of Washington
Dr. Mandy Chen (CoI)	Carnegie Institution of Washington
Jacob Charles Reinheimer (CoI)	University of North Texas
Prof. Sean Johnson (CoI)	University of Michigan
Dr. Ohad Shemmer (CoI)	University of North Texas

OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
Observation Folder				
	1		NIRCam Wide Field Slitless Spectroscopy	(1) J0439+1634AB
	2		NIRCam Wide Field Slitless Spectroscopy	(2) J0946+1835AB
	3		NIRCam Wide Field Slitless Spectroscopy	(4) J2037-4537AB
	4		NIRCam Wide Field Slitless Spectroscopy	(5) J2150-4652AB
	5		NIRCam Wide Field Slitless Spectroscopy	(3) J1452+4224AB

ABSTRACT

The first 2 to 3 Gyr of cosmic time saw the rise and peak of cosmic star formation, which was regulated by the interplay of gas accretion and outflows in galaxies. Observations of the circumgalactic medium (CGM) provide the most direct means to understand this baryon cycle. However,

traditional CGM observations are limited by the use of single background sightlines, which provide only a 1D pencil-beam view of individual halos. In contrast, multiply lensed QSOs provide a powerful means to spatially resolve the CGM in absorption. We propose the Lensed-QSO Absorption of the CGM at Early Stages (LACES) survey: a sensitive, flux-limited NIRCcam spectroscopic program targeting galaxies around five of the highest-redshift doubly imaged lensed QSOs ($z=4.1-6.5$). The resulting $R\sim 1500$ spectra and wide spectral coverage (2.5 to 4 micron) will efficiently detect >20 high-redshift ($z>2.7$) galaxies at $d<120$ kpc ($\sim 1.5-2 R_{\text{vir}}$) from these QSOs using strong nebular emission lines (e.g. H α and [OIII]), without target preselection bias. We will also obtain NIRCcam images (0.7 to 3.6 micron) to resolve the morphology and determine the stellar mass, star formation rate, and stellar population age of the galaxies. Using approved ground-based spectra, we will measure N(HI) and metal absorbers to determine CGM metallicities and elemental ratios, key for distinguishing pristine inflows from enriched outflows. With this program, we will move beyond traditional one-point samplings of the $z>2$ CGM, characterize CGM gas flows at high redshifts, map the spatial distribution of metals and its small-scale variations ($<\sim 100$ pc), and understand how the CGM depends on galaxy properties in early epochs.

OBSERVING DESCRIPTION

Using NIRCcam, we will perform slitless grism spectroscopy and multiband imaging of five high-redshift ($z=4.1-6.5$) gravitationally lensed QSO fields. The primary goals are to (1) identify and characterize high-redshift ($z>2.7$) galaxies in the foreground of these quasars, and (2) spatially resolve CGM gas flows around these galaxies to constrain their chemical enrichment and physical origins.

The LACES survey first consists of NIRCcam wide-field slitless spectroscopy (WFSS) using the F277W and F356W filters for continuous galaxy spectra between 2.5 and 4 microns. To disentangle overlapping spectra, we will use both Grisms R and C to disperse light in orthogonal directions, yielding an optimal FOV (area of full spectral coverage) of roughly $32''\times 32''$ (F277W) and $76''\times 76''$ (F356W), with partial spectral coverage expected over a significantly larger area. We aim to reach a 1-sigma emission line flux limit of 10^{-18} erg/s/cm 2 , requiring a total exposure time per field of 210 minutes in F277W and 124 minutes in F356W. To optimize data quality, we will use a 3-point INTRAMODULEX primary dither pattern combined with a 4-point subpixel dither, for a total of 12 dithered exposures per grism. We will use the recommended SHALLOW4 readout mode.

The observations also include concurrent imaging in four SW filters (F070W, F115W, F150W, F200W) and subsequent imaging in one LW filter (F356W). With total exposure times per field of 105, 105, 62, and 62 minutes for the F070W, F115W, F150W, and F200W filters, respectively, and 14 minutes for the F356W filter, we will detect faint ($m_{\text{AB}} \sim 26.5$, $0.05 L^*$) galaxies at $S/N>7-10$ in all filters.

Including overhead, the expected total charged time for our program is 50.6 hours.

Proposal 11567 - Targets - A High-Definition View of the Baryon Cycle in the First Two Billion Years

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	J0439+1634AB	RA: 04 39 47.0800 (69.9461667d) Dec: +16 34 15.70 (16.57103d) Equinox: J2000 Comments: Category=Galaxy Description=[Quasars] Extended=NO		
(2)	J0946+1835AB	RA: 09 46 4.9000 (146.5204167d) Dec: +18 35 41.80 (18.59494d) Equinox: J2000 Comments: Category=Galaxy Description=[Quasars] Extended=NO		
(3)	J1452+4224AB	RA: 14 52 11.5000 (223.0479167d) Dec: +42 24 29.02 (42.40806d) Equinox: J2000 Comments: Category=Galaxy Description=[Quasars] Extended=NO		
(4)	J2037-4537AB	RA: 20 37 21.2700 (309.3386250d) Dec: -45 37 48.80 (-45.63022d) Equinox: J2000 Comments: Category=Galaxy Description=[Quasars] Extended=NO		
(5)	J2150-4652AB	RA: 21 50 28.7100 (327.6196250d) Dec: -46 52 51.30 (-46.88092d) Equinox: J2000 Comments: Category=Galaxy Description=[Quasars] Extended=NO		

Proposal 11567 - Observation 1 - A High-Definition View of the Baryon Cycle in the First Two Billion Years

Observation	Proposal 11567, Observation 1											Tue Mar 17 15:00:16 GMT 2026
	Diagnostic Status: Warning											
	Observing Template: NIRCam Wide Field Slitless Spectroscopy											
Diagnostics	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(1)	J0439+1634AB	RA: 04 39 47.0800 (69.9461667d) Dec: +16 34 15.70 (16.57103d) Equinox: J2000									
	Comments: Category=Galaxy Description=[Quasars] Extended=NO											
Template	Module				Subarray				Grism (Long Wavelength)			
	A				FULL				BOTH			
Dithers	#	Primary Dither Type				Primary Dithers				Subpixel Positions		
	1	INTRAMODULEX				3				2-Point		
Direct Image	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID	Grism (Long Wavelength)	Exposure Type	Total Dithers
	1	F115W	F356W	SHALLOW4	4	1	1	203.999	274273.7	GRISMR	Direct Image	1
	2	F200W	F356W	SHALLOW4	4	1	1	203.999	274273.7	GRISMR	Direct Image	1
	3	F115W	F356W	SHALLOW4	4	1	1	203.999	274273.7	GRISMC	Direct Image	1
	4	F200W	F356W	SHALLOW4	4	1	1	203.999	274273.7	GRISMC	Direct Image	1

Proposal 11567 - Observation 1 - A High-Definition View of the Baryon Cycle in the First Two Billion Years

Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID	Grism (Long Wavelength)	Exposure Type	Total Dithers
	1	F070W	F277W	SHALLOW4	10	1	6	3156.61	274273.33	GRISMR	Grism (Long Wavelength)	6
	2	F115W	F277W	SHALLOW4	10	1	6	3156.61	274273.33	GRISMR	Grism (Long Wavelength)	6
	3	F150W	F356W	SHALLOW4	6	1	6	1868.198	274273.27	GRISMR	Grism (Long Wavelength)	6
	4	F200W	F356W	SHALLOW4	6	1	6	1868.198	274273.27	GRISMR	Grism (Long Wavelength)	6
	5	F200W	F356W	SHALLOW4	4	1	2	407.997	274273.7		Out of Field	2
	6	F070W	F277W	SHALLOW4	10	1	6	3156.61	274273.33	GRISMC	Grism (Long Wavelength)	6
	7	F115W	F277W	SHALLOW4	10	1	6	3156.61	274273.33	GRISMC	Grism (Long Wavelength)	6
	8	F150W	F356W	SHALLOW4	6	1	6	1868.198	274273.27	GRISMC	Grism (Long Wavelength)	6
	9	F200W	F356W	SHALLOW4	6	1	6	1868.198	274273.27	GRISMC	Grism (Long Wavelength)	6
	10	F200W	F356W	SHALLOW4	4	1	2	407.997	274273.7		Out of Field	2

Proposal 11567 - Observation 2 - A High-Definition View of the Baryon Cycle in the First Two Billion Years

Observation	Proposal 11567, Observation 2										Tue Mar 17 15:00:16 GMT 2026	
	Diagnostic Status: Warning											
	Observing Template: NIRCam Wide Field Slitless Spectroscopy											
Diagnostics	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
	(Visit 2:2) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
	(Visit 2:3) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(2)	J0946+1835AB	RA: 09 46 4.9000 (146.5204167d) Dec: +18 35 41.80 (18.59494d) Equinox: J2000									
	Comments: Category=Galaxy Description=[Quasars] Extended=NO											
Template	Module				Subarray				Grism (Long Wavelength)			
	A				FULL				BOTH			
Dithers	#	Primary Dither Type				Primary Dithers				Subpixel Positions		
	1	INTRAMODULEX				3				2-Point		
Direct Image	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID	Grism (Long Wavelength)	Exposure Type	Total Dithers
	1	F115W	F356W	SHALLOW4	4	1	1	203.999	274273.7	GRISMR	Direct Image	1
	2	F200W	F356W	SHALLOW4	4	1	1	203.999	274273.7	GRISMR	Direct Image	1
	3	F115W	F356W	SHALLOW4	4	1	1	203.999	274273.7	GRISMC	Direct Image	1
	4	F200W	F356W	SHALLOW4	4	1	1	203.999	274273.7	GRISMC	Direct Image	1

Proposal 11567 - Observation 2 - A High-Definition View of the Baryon Cycle in the First Two Billion Years

	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID	Grism (Long Wavelength)	Exposure Type	Total Dithers
Spectral Elements	1	F070W	F277W	SHALLOW4	10	1	6	3156.61	274273.33	GRISMR	Grism (Long Wavelength)	6
	2	F115W	F277W	SHALLOW4	10	1	6	3156.61	274273.33	GRISMR	Grism (Long Wavelength)	6
	3	F150W	F356W	SHALLOW4	6	1	6	1868.198	274273.27	GRISMR	Grism (Long Wavelength)	6
	4	F200W	F356W	SHALLOW4	6	1	6	1868.198	274273.27	GRISMR	Grism (Long Wavelength)	6
	5	F200W	F356W	SHALLOW4	4	1	2	407.997	274273.7		Out of Field	2
	6	F070W	F277W	SHALLOW4	10	1	6	3156.61	274273.33	GRISMC	Grism (Long Wavelength)	6
	7	F115W	F277W	SHALLOW4	10	1	6	3156.61	274273.33	GRISMC	Grism (Long Wavelength)	6
	8	F150W	F356W	SHALLOW4	6	1	6	1868.198	274273.27	GRISMC	Grism (Long Wavelength)	6
	9	F200W	F356W	SHALLOW4	6	1	6	1868.198	274273.27	GRISMC	Grism (Long Wavelength)	6
	10	F200W	F356W	SHALLOW4	4	1	2	407.997	274273.7		Out of Field	2
Special Requirements	Sequence Visits within 53.0 Days Visits Same PA											

Proposal 11567 - Observation 3 - A High-Definition View of the Baryon Cycle in the First Two Billion Years

Observation	Proposal 11567, Observation 3										Tue Mar 17 15:00:16 GMT 2026	
	Diagnostic Status: Warning											
	Observing Template: NIRCam Wide Field Slitless Spectroscopy											
Diagnostics	(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(4)	J2037-4537AB	RA: 20 37 21.2700 (309.3386250d) Dec: -45 37 48.80 (-45.63022d) Equinox: J2000									
	Comments: Category=Galaxy Description=[Quasars] Extended=NO											
Template	Module				Subarray				Grism (Long Wavelength)			
	A				FULL				BOTH			
Dithers	#	Primary Dither Type				Primary Dithers				Subpixel Positions		
	1	INTRAMODULEX				3				2-Point		
Direct Image	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID	Grism (Long Wavelength)	Exposure Type	Total Dithers
	1	F115W	F356W	SHALLOW4	4	1	1	203.999	274273.7	GRISMR	Direct Image	1
	2	F200W	F356W	SHALLOW4	4	1	1	203.999	274273.7	GRISMR	Direct Image	1
	3	F115W	F356W	SHALLOW4	4	1	1	203.999	274273.7	GRISMC	Direct Image	1
	4	F200W	F356W	SHALLOW4	4	1	1	203.999	274273.7	GRISMC	Direct Image	1

Proposal 11567 - Observation 3 - A High-Definition View of the Baryon Cycle in the First Two Billion Years

Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID	Grism (Long Wavelength)	Exposure Type	Total Dithers
	1	F070W	F277W	SHALLOW4	10	1	6	3156.61	274273.33	GRISMR	Grism (Long Wavelength)	6
	2	F115W	F277W	SHALLOW4	10	1	6	3156.61	274273.33	GRISMR	Grism (Long Wavelength)	6
	3	F150W	F356W	SHALLOW4	6	1	6	1868.198	274273.27	GRISMR	Grism (Long Wavelength)	6
	4	F200W	F356W	SHALLOW4	6	1	6	1868.198	274273.27	GRISMR	Grism (Long Wavelength)	6
	5	F200W	F356W	SHALLOW4	4	1	2	407.997	274273.7		Out of Field	2
	6	F070W	F277W	SHALLOW4	10	1	6	3156.61	274273.33	GRISMC	Grism (Long Wavelength)	6
	7	F115W	F277W	SHALLOW4	10	1	6	3156.61	274273.33	GRISMC	Grism (Long Wavelength)	6
	8	F150W	F356W	SHALLOW4	6	1	6	1868.198	274273.27	GRISMC	Grism (Long Wavelength)	6
	9	F200W	F356W	SHALLOW4	6	1	6	1868.198	274273.27	GRISMC	Grism (Long Wavelength)	6
	10	F200W	F356W	SHALLOW4	4	1	2	407.997	274273.7		Out of Field	2

Proposal 11567 - Observation 4 - A High-Definition View of the Baryon Cycle in the First Two Billion Years

Observation	Proposal 11567, Observation 4										Tue Mar 17 15:00:16 GMT 2026	
	Diagnostic Status: Warning											
	Observing Template: NIRCam Wide Field Slitless Spectroscopy											
Diagnostics	(Visit 4:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
	(Visit 4:2) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
	(Visit 4:3) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(5)	J2150-4652AB	RA: 21 50 28.7100 (327.6196250d) Dec: -46 52 51.30 (-46.88092d) Equinox: J2000									
	Comments: Category=Galaxy Description=[Quasars] Extended=NO											
Template	Module				Subarray				Grism (Long Wavelength)			
	A				FULL				BOTH			
Dithers	#	Primary Dither Type				Primary Dithers				Subpixel Positions		
	1	INTRAMODULEX				3				2-Point		
Direct Image	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID	Grism (Long Wavelength)	Exposure Type	Total Dithers
	1	F115W	F356W	SHALLOW4	4	1	1	203.999	274273.7	GRISMR	Direct Image	1
	2	F200W	F356W	SHALLOW4	4	1	1	203.999	274273.7	GRISMR	Direct Image	1
	3	F115W	F356W	SHALLOW4	4	1	1	203.999	274273.7	GRISMC	Direct Image	1
	4	F200W	F356W	SHALLOW4	4	1	1	203.999	274273.7	GRISMC	Direct Image	1

Proposal 11567 - Observation 4 - A High-Definition View of the Baryon Cycle in the First Two Billion Years

Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID	Grism (Long Wavelength)	Exposure Type	Total Dithers
	1	F070W	F277W	SHALLOW4	10	1	6	3156.61	274273.33	GRISMR	Grism (Long Wavelength)	6
	2	F115W	F277W	SHALLOW4	10	1	6	3156.61	274273.33	GRISMR	Grism (Long Wavelength)	6
	3	F150W	F356W	SHALLOW4	6	1	6	1868.198	274273.27	GRISMR	Grism (Long Wavelength)	6
	4	F200W	F356W	SHALLOW4	6	1	6	1868.198	274273.27	GRISMR	Grism (Long Wavelength)	6
	5	F200W	F356W	SHALLOW4	4	1	2	407.997	274273.7		Out of Field	2
	6	F070W	F277W	SHALLOW4	10	1	6	3156.61	274273.33	GRISMC	Grism (Long Wavelength)	6
	7	F115W	F277W	SHALLOW4	10	1	6	3156.61	274273.33	GRISMC	Grism (Long Wavelength)	6
	8	F150W	F356W	SHALLOW4	6	1	6	1868.198	274273.27	GRISMC	Grism (Long Wavelength)	6
	9	F200W	F356W	SHALLOW4	6	1	6	1868.198	274273.27	GRISMC	Grism (Long Wavelength)	6
	10	F200W	F356W	SHALLOW4	4	1	2	407.997	274273.7		Out of Field	2
Special Requirements	Sequence Visits within 53.0 Days Visits Same PA											

Proposal 11567 - Observation 5 - A High-Definition View of the Baryon Cycle in the First Two Billion Years

Observation	Proposal 11567, Observation 5											Tue Mar 17 15:00:16 GMT 2026
	Diagnostic Status: Warning											
	Observing Template: NIRCam Wide Field Slitless Spectroscopy											
Diagnostics	(Visit 5:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
	(Visit 5:2) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
	(Visit 5:3) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(3)	J1452+4224AB	RA: 14 52 11.5000 (223.0479167d) Dec: +42 24 29.02 (42.40806d) Equinox: J2000									
	Comments: Category=Galaxy Description=[Quasars] Extended=NO											
Template	Module				Subarray				Grism (Long Wavelength)			
	A				FULL				BOTH			
Dithers	#	Primary Dither Type				Primary Dithers				Subpixel Positions		
	1	INTRAMODULEX				3				2-Point		
Direct Image	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID	Grism (Long Wavelength)	Exposure Type	Total Dithers
	1	F115W	F356W	SHALLOW4	4	1	1	203.999	274273.7	GRISMR	Direct Image	1
	2	F200W	F356W	SHALLOW4	4	1	1	203.999	274273.7	GRISMR	Direct Image	1
	3	F115W	F356W	SHALLOW4	4	1	1	203.999	274273.7	GRISMC	Direct Image	1
	4	F200W	F356W	SHALLOW4	4	1	1	203.999	274273.7	GRISMC	Direct Image	1

Proposal 11567 - Observation 5 - A High-Definition View of the Baryon Cycle in the First Two Billion Years

	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID	Grism (Long Wavelength)	Exposure Type	Total Dithers
Spectral Elements	1	F070W	F277W	SHALLOW4	10	1	6	3156.61	274273.33	GRISMR	Grism (Long Wavelength)	6
	2	F115W	F277W	SHALLOW4	10	1	6	3156.61	274273.33	GRISMR	Grism (Long Wavelength)	6
	3	F150W	F356W	SHALLOW4	6	1	6	1868.198	274273.27	GRISMR	Grism (Long Wavelength)	6
	4	F200W	F356W	SHALLOW4	6	1	6	1868.198	274273.27	GRISMR	Grism (Long Wavelength)	6
	5	F200W	F356W	SHALLOW4	4	1	2	407.997	274273.7		Out of Field	2
	6	F070W	F277W	SHALLOW4	10	1	6	3156.61	274273.33	GRISMC	Grism (Long Wavelength)	6
	7	F115W	F277W	SHALLOW4	10	1	6	3156.61	274273.33	GRISMC	Grism (Long Wavelength)	6
	8	F150W	F356W	SHALLOW4	6	1	6	1868.198	274273.27	GRISMC	Grism (Long Wavelength)	6
	9	F200W	F356W	SHALLOW4	6	1	6	1868.198	274273.27	GRISMC	Grism (Long Wavelength)	6
	10	F200W	F356W	SHALLOW4	4	1	2	407.997	274273.7		Out of Field	2
Special Requirements	Sequence Visits within 53.0 Days Visits Same PA											